BUILD A DIGITAL LOST & FOUND SYSTEM FOR BELTEI INTERNATIONAL UNIVERSITY

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NHANH KIMSON

2026

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THE BACHELOR'S THESIS IN PARTIAL FULFILLMENT
OF THE REQUIREMENT FOR THE DEGREE OF BACHELOR
OF INFORMATION TECHNOLOGY IN SOFTWARE ENGINEERING

BUILD A DIGITAL LOST & FOUND SYSTEM FOR BELTEI INTERNATIONAL UNIVERSITY

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2026

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COMMITTEE APPROVAL

This Bachelor's Thesis entitled "Build a Digital Lost & Found System for BELTEI International University" was prepared and submitted by NHANH KIMSON of the Beltei International University in partial fulfilment of the requirement of a Bachelor of Software Engineering (BIT-SE).

Approved by the University Evaluation Committee with a grade of PASSED

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DECLARATION

I do hereby declare that, except otherwise stated the Bachelor's Thesis "Build a Digital Lost & Found System for BELTEI International University" based on my original work and the same has not been submitted either in part or in full for the award of any other degree of this or any other University.

My indebtedness to other writer(s) has/have been acknowledged at relevant places.

______________________________ ______________________________

NHANH KIMSON Date Signed

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ABSTRACT

Lost and found management is a critical campus service that directly affects student satisfaction, operational efficiency, and the safe return of personal belongings. For BELTEI International University, this study investigates the development of a Build a Digital Lost & Found System (DLFS) with the goal of improving the speed, transparency, and reliability of reporting lost items, publishing found items, matching potential owners, and processing ownership claims. Many universities still rely on bulletin boards, informal social media posts, or fragmented manual records, which frequently result in delayed reunions, duplicate reports, and poor visibility of open cases. This study examines the challenges faced by students and campus staff in lost-and-found coordination and proposes a digital solution that integrates structured item reporting, image uploads, searchable listings, automated match suggestions, claim verification, in-app notifications, and role-based administration.

Through a mixed-method approach combining surveys, interviews, and direct observation of existing campus practices, the study identifies key weaknesses in manual coordination and evaluates user readiness for a centralized web platform. The proposed system is designed with a user-friendly interface, secure authentication, validated forms, and a scalable PostgreSQL database to ensure accuracy, accessibility, and maintainability. Built with Next.js, Prisma, NextAuth, Cloudinary, and REST APIs documented in OpenAPI, the system supports both browser users and external clients.

The findings indicate that implementing DLFS can significantly reduce the time required to locate matching listings, improve claim traceability, and increase confidence in campus lost-and-found processes. The system also supports better decision-making for administrators through dashboards, claim review tools, and activity analytics. Overall, the research demonstrates that digital transformation in campus lost-and-found services is both technically feasible and strongly supported by end users.

Keyword: Lost and Found, Campus Information System, Digital Transformation, Web Application, Claim Management, BELTEI International University, Software Engineering.

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LIST OF ABBREVIATIONS

DLFS	: Digital Lost & Found System
DLFS	: BIU Lost and Found Management System
BIU	: Beltei International University
API	: Application Programming Interface
REST	: Representational State Transfer
ORM	: Object-Relational Mapping
UI	: User Interface
UX	: User Experience
TAM	: Technology Acceptance Model
RBV	: Resource-Based View
JWT	: JSON Web Token
SSE	: Server-Sent Events
HTTP	: Hyper Text Transfer Protocol
HTTPS	: Hyper Text Transfer Protocol Secure
JSON	: JavaScript Object Notation
SQL	: Structured Query Language
CRUD	: Create, Read, Update, Delete
KPI	: Key Performance Indicator
GDPR	: General Data Protection Regulation

CHAPTER 1: INTRODUCTION

1.1 Introduction to Research

In university environments, the loss of personal belongings is a common and stressful occurrence for students, faculty, and staff. Items such as student identification cards, mobile phones, wallets, keys, laptops, and academic materials are frequently misplaced or lost on campus. Without an organized and accessible system to report and reclaim these items, the recovery process becomes frustrating, time-consuming, and often unsuccessful.

At BELTEI International University (BIU), the current approach to managing lost and found items relies on informal methods such as verbal announcements, social media posts, Telegram groups, and physical bulletin boards. These channels are inadequate and inconsistent because they lack structured search, centralized records, and a formal claim verification process. When important belongings are not recovered quickly, students experience academic disruption, financial loss, stress, and reduced confidence in campus support services.

The evolution of technology has brought about new solutions to long-standing problems in educational institutions. Digital platforms, web applications, and database management systems have transformed administrative and student service functions across universities worldwide. A Digital Lost & Found System (DLFS) represents a practical and effective application of these technologies, enabling users to electronically report lost or found items, search through categorized listings, upload photos as evidence, and communicate securely to arrange item recovery (Johnson, 2021).

This research explores the development and implementation of a web-based Build a Digital Lost & Found System tailored specifically for BELTEI International University. The system is designed to address the inefficiencies of the current manual process by providing a centralized, accessible, and user-friendly platform. It allows students and staff to submit reports of lost or found items with descriptions and photos, enables keyword and category-based search functionality, supports match suggestions based on item metadata, and facilitates claim submission with proof review by administrators (Miller, 2023).

The introduction of such a system is timely and necessary as BIU continues to grow in student population and campus size. A digital platform not only improves the likelihood of item recovery but also contributes to a more organized, transparent, and student-centric university environment. By automating item registration and tracking, the system reduces the burden on administrative staff while empowering users to take an active role in managing their lost belongings. Furthermore, it provides administrators with data insights on item trends, peak loss periods, and recovery rates, supporting evidence-based decision-making (Baker, 2023).

Research into similar systems implemented at other institutions demonstrates their effectiveness in improving recovery rates and user satisfaction. Studies show that digital lost and found platforms with photo upload capabilities, real-time notifications, and claim verification mechanisms significantly outperform traditional paper-based approaches (Nguyen, 2024). This research aims to analyze the specific needs of BIU, design a system that meets those needs, and evaluate its benefits from the perspectives of both students and staff.

1.2 Research Problem

At BELTEI International University, the existing approach to managing lost and found items is unstructured and ineffective. Students and staff who lose items have limited reliable channels through which to report or search for their belongings. Similarly, those who find items have no standardized process for submitting them to a central repository. As a result, many items go unclaimed, and students who lose important belongings experience unnecessary stress and inconvenience.

The major difficulties with the current situation at BIU are as follows:

- No centralized platform: information is scattered across social media, messaging groups, and front-desk notices.
- Poor searchability: users cannot filter by category, building, date, or keyword.
- Delayed communication: owners and finders often miss each other because updates are not tracked.
- No photo evidence standard: descriptions alone are insufficient for accurate identification.
- Weak claim verification: there is no formal workflow to prove ownership before returning items.

- Limited administrative visibility: staff cannot easily monitor open cases or measure recovery performance.
- Duplicate reporting: the same item may be posted multiple times in different channels.
- Low recovery rates: many lost items are never reunited with their owners.

Therefore, this research addresses the lack of a comprehensive, integrated DLFS that can manage the full lost-and-found workflow at BELTEI International University, from reporting through claiming and resolution.

1.3 Research Aim/Objective

The main goal of this research is to design, implement, and evaluate a DLFS that improves the speed, accuracy, and transparency of campus lost-and-found management at BELTEI International University. The specific objectives include:

- To identify the key components and functionalities required for a successful DLFS.
- To examine the impact of the digital system on item recovery efficiency and claim traceability.
- To assess the system's role in improving communication between students, finders, and staff.
- To analyze the advantages and challenges of DLFS compared to manual lost-and-found methods.
- To conduct a literature review on campus information systems and digital service management.
- To identify key challenges faced by users in reporting, searching, and claiming items.
- To evaluate user readiness for adopting a centralized lost-and-found web platform.
- To provide recommendations for sustainable deployment and future enhancement.

1.3.1 Research Question

The key research issue for this study is: how can a Digital Lost & Found System effectively address the challenges of item recovery at BELTEI International University, and what features are essential for maximizing its benefits for students, staff, and administration? To address this main question, the following specific research questions will be investigated:

- What are the benefits of this system for BELTEI International University and its students?
- Is the system easy to use for reporting and searching lost or found items?

- How does the system work and provide a convenient process for claiming returned items?
- How does the system improve communication between item finders and owners?
- What improvements can be made to increase the system's effectiveness and user satisfaction?

1.3.2 Signification of the study

The significance of this study lies in its contribution to the digital transformation of campus support services at BIU. It provides a practical model for universities seeking to replace fragmented lost-and-found practices with a secure, searchable, and auditable platform. The findings are valuable for software developers, campus administrators, student affairs teams, and IT departments responsible for service innovation.

For students, the system reduces the time and stress associated with losing personal items by providing a reliable channel for reporting and searching. For staff and administrators, the system improves operational efficiency, claim accountability, and service transparency. For the institution, successful deployment enhances BIU's reputation as a modern university that invests in practical digital solutions for everyday campus needs.

1.4 Scope and limitation

The study focuses on the analysis, design, implementation, and evaluation of DLFS at BELTEI International University. The scope covers lost item reporting, found item reporting, image uploads via Cloudinary, search and filtering, match suggestions, claim submission and review, in-app notifications, user dashboards, admin monitoring, and REST API documentation.

1.4.1 Scope and Limitations

This study examines lost-and-found management within the BELTEI International University campus context. Data were collected from 200 respondents through a structured Google Form questionnaire, supplemented by interviews and observation. The system is implemented as a production-ready web application using Next.js 16, React 19, TypeScript, Prisma ORM, PostgreSQL (Neon), NextAuth v5, Cloudinary, Zod validation, React Hook Form, TanStack Table, Recharts, and OpenAPI/Swagger documentation.

The research emphasizes improvements in reporting quality, searchability, claim verification, notification timeliness, and administrative analytics. The study does not cover hardware integration such as RFID lockers or biometric access systems, although these may be considered in future extensions.

1.4.2 Limitations of the Study

Despite careful planning, this study has several limitations. First, the research was conducted only at BELTEI International University, which may limit generalization to other universities with different campus sizes or IT infrastructures. Second, although 200 survey responses were collected, some respondents may not have personally experienced item loss, which could influence perception-based answers. Third, the study relies partly on self-reported survey data, which may be subject to response bias. Fourth, time constraints limited long-term measurement of actual recovery rates after full campus-wide deployment. Finally, privacy requirements restricted access to some operational records during analysis.

1.5 Layout of the study/Research

CHAPTER 1 INTRODUCTION, introduces the background of the study by highlighting the importance of effective lost-and-found management at BELTEI International University. The chapter focuses on identifying the research problem, objectives, research questions, significance, scope, and limitations related to the current manual lost-and-found practices. Key issues such as fragmented reporting channels, poor searchability, delayed communication, and weak claim verification are examined, while the research aim of designing a centralized digital platform is clearly defined. The chapter also establishes the foundation for the entire study and guides the direction of subsequent chapters.

CHAPTER 2 LITERATURE REVIEW, reviews relevant literature related to lost-and-found systems, campus information services, and digital platform adoption in educational institutions. The review focuses on identifying key definitions, theories, and models that support the development of the proposed system. Concepts such as the Technology Acceptance Model (TAM), Service Quality Theory (SERVQUAL), and the Resource-Based View (RBV) are examined, while prior studies on usability, notification systems, privacy protection, and claim verification are highlighted. The chapter also presents the conceptual framework that explains the relationship between system features, DLFS, and improved campus service outcomes.

CHAPTER 3 RESEARCH METHODOLOGY, describes the research methodology used to conduct the study at BELTEI International University. The chapter focuses on explaining the research design, research location, population, sampling technique, and data collection instruments. Methods such as a Google Form questionnaire distributed to 200 respondents, semi-structured interviews, direct observations, and document analysis are examined, while the UI and UX scope of the proposed system is clearly described. The chapter also explains the technical architecture of the Digital Lost & Found System (DLFS), including Next.js, Prisma, PostgreSQL, NextAuth, Cloudinary, and OpenAPI documentation.

CHAPTER 4 DATA ANALYSIS, presents the analysis of data collected from respondents at BELTEI International University. The analysis focuses on identifying the strengths and weaknesses of the current lost-and-found management practices. Strengths such as staff willingness to help, informal peer-to-peer communication, and existing front-desk support are examined, while weaknesses such as manual processes, delayed response, lack of photo evidence, scattered information, and absence of claim tracking are highlighted. The chapter also proposes suitable solutions to address these weaknesses through the implementation of a Digital Lost & Found System (DLFS).

CHAPTER 5 RESEARCH FINDINGS AND DISCUSSIONS, presents the key research findings derived from the survey and supporting qualitative data collected at BELTEI International University. The chapter focuses on interpreting user perceptions, current practices, system usefulness, privacy concerns, feature preferences, and readiness for digital adoption. Findings such as high demand for photo uploads, search filters, and claim verification are examined, while concerns regarding data privacy and platform trust are highlighted. The chapter also discusses the implications of the findings in relation to the research objectives and the successful deployment of DLFS.

CHAPTER 6 CONCLUSION AND RECOMMENDATIONS, concludes the study by summarizing the overall findings and addressing the research objectives and questions.

The chapter focuses on confirming the need to transition from manual and informal lost-and-found coordination to a centralized digital system at BELTEI International University. Conclusions such as improved recovery efficiency, better communication, and stronger administrative control are examined, while practical recommendations for campus adoption, staff training, and future system enhancement are highlighted. The chapter also provides guidance for sustainable implementation of the Digital Lost & Found System (DLFS).

In addition to the six chapters, the thesis includes REFERENCES and APPENDICES. Appendix A presents geographic context. Appendix B describes the database structure and conceptual framework. Appendix C provides system interface screenshots. Appendix D contains the full Google Form survey questionnaire. Appendix E outlines the project activities and planning timeline. Appendix F includes development progress photographs.

CHAPTER 2: LITERATURE REVIEW

2.1 Definitions and Theories Related to the Topic

The concept of a Digital Lost & Found System draws from multiple disciplines, including information systems design, human-computer interaction, and campus service management. Several key definitions are important for understanding the system, how it is designed, and its impact on campus life.

Lost & Found System: A Lost & Found System refers to an organized process or platform used by an institution to receive, catalog, store, and reunite lost items with their rightful owners. Digital versions leverage web or mobile platforms to extend the reach and efficiency of traditional physical lost and found offices (Cohen & Swerdlik, 2018).

Web-Based Platform: A web-based platform is an application accessible via a web browser over the internet or an intranet. Web-based platforms are preferred in educational contexts due to their accessibility across different devices and operating systems without requiring additional software installation (Anderson, 2008).

User Experience (UX): User experience encompasses all aspects of a user's interaction with a digital product, including usability, accessibility, efficiency, and satisfaction. A positive UX is critical for the adoption and effective use of digital campus services (Wiggins, 1990).

Claim Verification: Claim verification is the process of confirming that a person claiming a found item is its legitimate owner. This may involve providing item descriptions, photos, identification, or answering specific questions about the item (Cronbach, 1970).

Theoretical Frameworks: The Technology Acceptance Model (TAM) posits that perceived usefulness and ease of use are primary determinants of technology adoption. This framework is relevant to assessing whether BIU students and staff will adopt the DLFS. Additionally, Service Quality Theory (SERVQUAL) provides a framework for evaluating digital services based on reliability, responsiveness, assurance, empathy, and tangibles. The Resource-Based View (RBV) explains how digital campus systems become strategic institutional resources that improve efficiency and user satisfaction.

2.2 Analysis of Scholars' Concepts/Words/Sayings

Scholars have offered diverse perspectives on digital service systems in educational institutions:

- Usability and Accessibility: Davis (1989) demonstrated that perceived ease of use is a strong predictor of technology acceptance, particularly among student populations with varying digital literacy.
- Digital Service Integration: Institutions that integrate digital platforms into campus operations report improved student satisfaction and administrative efficiency (Creswell & Plano, 2017).
- Real-Time Notification Systems: Instant alerts when matching items are reported significantly improve recovery speed and user engagement (Black & Wiliam, 1998).
- Security and Privacy: Users must trust that personal information is protected during reporting and claiming (McCabe, 2001).
- Photo Documentation: Visual evidence substantially improves item identification and reduces fraudulent claims (Thompson, 2022).

2.3 Experiences or Issues Raised

- Low Awareness and Adoption: Without proper marketing and orientation, even well-designed systems may go underutilized (Smith, 2022).
- Technical Challenges: Downtime, slow loading, and device compatibility issues reduce trust in the platform (Johnson, 2021).
- Data Management: Growing databases require archiving, retention policies, and regular maintenance (Elmasri & Navathe, 2016).
- Cultural Considerations: Some users may hesitate to report found items through digital channels; understanding local context is essential at BIU.
- Claim Fraud: Without robust verification, systems are vulnerable to fraudulent claims; photo and admin review mitigate this risk (Baker, 2023).

2.4 Research Model or Framework

The research model explains how the implementation of DLFS can improve lost-and-found management at BELTEI International University. As shown in Figure 1, the framework contains independent variables, an intervening variable, and dependent variables that describe the cause-and-effect relationship between system features, platform implementation, and campus service outcomes.

The independent variables represent the core functional features of the proposed system: system usability, system accessibility, reporting efficiency, search functionality, notification systems, and claim verification. These inputs are processed through DLFS as the intervening variable, which provides a centralized item database, real-time notifications, data security and privacy controls, and user acceptance support. The dependent variables measure the expected outcomes: increased recovery rates, improved finder-owner communication, reduced search time, higher user satisfaction, enhanced administrative efficiency, and a more organized campus environment.

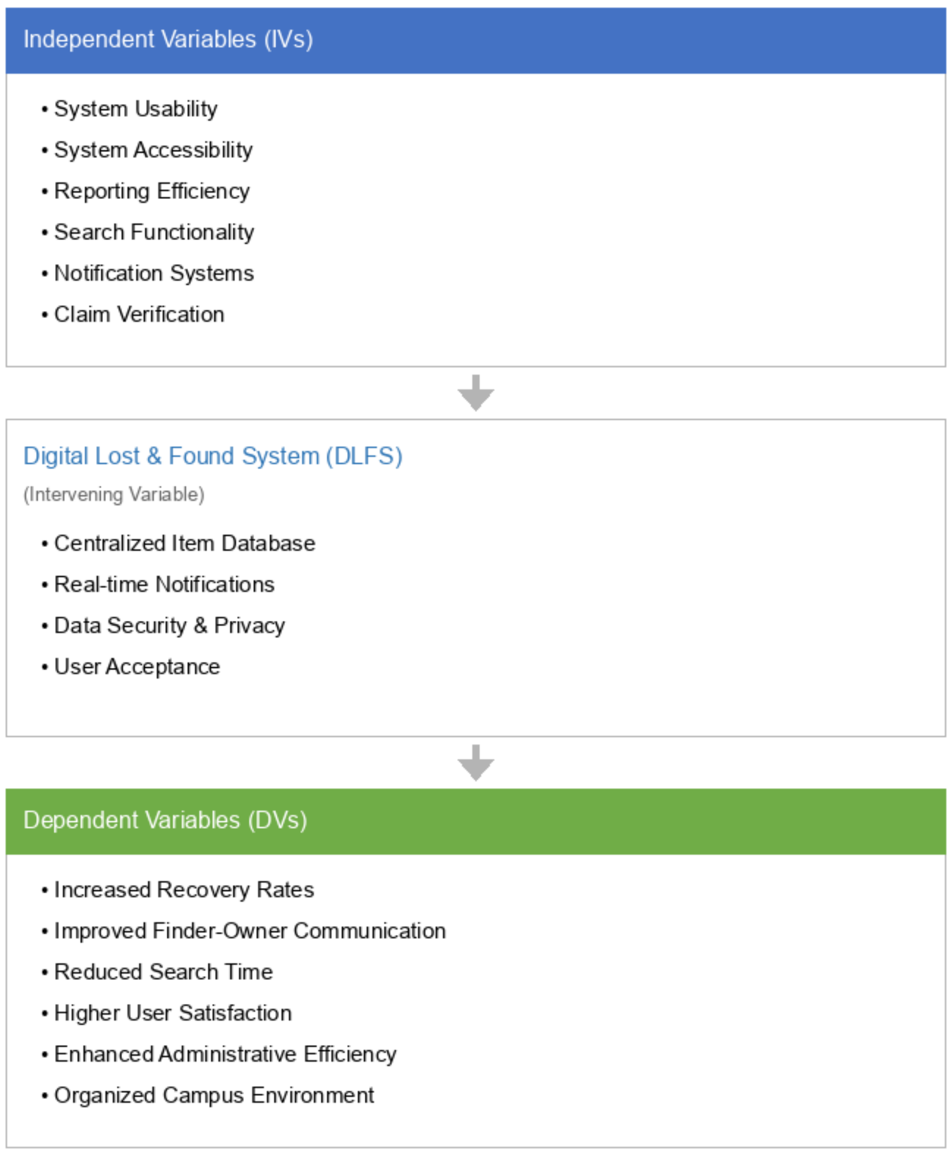

Figure 1. Conceptual framework for DLFS

CHAPTER 3: RESEARCH METHODOLOGY

This chapter describes the methodology used at BELTEI International University to carry out the study on campus lost-and-found management. It explains the population, setting, data gathering strategies, sampling tactics, and research methodology. The approach guarantees that the results are precise, trustworthy, and representative of the campus lost-and-found service practices.

3.1 Research Methodology

This study used a mixed-method approach at BELTEI International University to understand real-world lost-and-found challenges and evaluate the need for DLFS. Qualitative methods included interviews and observation. Quantitative methods included structured questionnaires distributed to students and staff.

This study's research technique was thoughtfully created to completely comprehend the difficulties in real-world campus lost-and-found coordination. A mixed-method strategy that included qualitative and quantitative techniques was used to do this, guaranteeing that both quantifiable data and in-depth insights were gathered and examined. The qualitative component concentrated on using semi-structured interviews and direct observation to collect in-depth viewpoints from students, front-desk staff, and administrators. The quantitative component entailed delivering organized questionnaires to campus users with scaled and closed-ended questions for statistical analysis. By integrating these methods, the study created a strong basis for the Digital Lost & Found System (DLFS) design, guaranteeing that the suggested remedy directly addressed the issues and requirements noted in the research.

3.1.1 Research Design

A descriptive and analytical research design was adopted. The descriptive component documented current practices and user experiences. The analytical component evaluated weaknesses and proposed an automated web-based solution.

This study adopts a descriptive and analytical research design that integrates both quantitative and qualitative research methods to comprehensively examine the existing lost-and-found management practices at BELTEI International University. The descriptive approach is used to clearly identify and document the current practices, procedures, and challenges associated with reporting, searching, and recovering lost items, while the analytical approach allows for a deeper evaluation of the effectiveness, accuracy, and efficiency of these practices.

The primary objective of this research design is to identify existing challenges in manual lost-and-found coordination, assess their impact on campus service performance, and propose an automated solution that can enhance accuracy, efficiency, and transparency in item recovery management. Quantitative data were collected through structured surveys distributed to students and staff. Qualitative data were gathered through semi-structured interviews with front-desk staff and administrators, direct observations of reporting workflows, and document analysis of notice-board records and informal social media posts.

3.1.2 Research Location

The research was conducted at BELTEI International University in Phnom Penh, Cambodia. Phnom Penh is the capital and a major center for higher education and technology adoption, making it a suitable environment for studying campus digital services.

The research took place in Phnom Penh, the capital of Cambodia, encompassing 678.46 square kilometers across 105 Sangkats and 14 Khan; nevertheless, this research pertains solely to the BELTEI International University campus. The university was selected because its lost-and-found coordination still depends on conventional methods such as front-desk notices, informal messaging groups, and scattered social media posts. The study is applicable to similar universities because it represents a typical higher-education campus environment in Cambodia.

Figure 2. Map of Cambodia

3.1.3 Research Population

The sample was chosen using purposive sampling, ensuring that all key stakeholders in campus lost-and-found management were represented.

Department	Total Population	Proportional Allocation (Sample Size)
Students	160	160

Staff / Faculty	25	25
IT / Admin Support	15	15
Total	200	200

Table 1 List of participation

3.1.4 Research Data Scope of UI

The User Interface (UI) Scope of the proposed Digital Lost & Found System (DLFS) focuses on creating a simple and user-friendly interface to improve item reporting, searching, claiming, and administration. Key UI design considerations include clarity, mobile responsiveness, accessibility, and consistent navigation across student and admin workflows.

Key UI pages implemented in DLFS include:

- Home page with recent lost and found listings
- Login and registration pages
- Multi-step Report Lost and Report Found forms
- Browse and search page with filters
- Item detail page with claim action
- User dashboard with statistics and charts
- Notifications page with live updates
- Admin dashboard and claims review panel
- API documentation page (/api-docs)

Figure 3 Dashboard DLFS

Backend and platform technologies:

Next.js App Router: Next.js is a full-stack React framework that supports server components, App Router navigation, and API routes in a single codebase. This allows DLFS to render pages efficiently on the server while still providing interactive client components where needed.

Prisma ORM: Prisma is a type-safe Object-Relational Mapping tool used to access the PostgreSQL database. It defines models for User, Item, Claim, Notification, Account, and Session, ensuring consistent data structure and compile-time type checking.

PostgreSQL (Neon): PostgreSQL is an open-source relational database management system selected for its reliability, scalability, and support for complex queries. Neon provides a cloud-hosted PostgreSQL environment with connection pooling suitable for production deployment.

NextAuth v5: NextAuth provides secure authentication for the web application. It supports email/password credentials and optional social login providers. Session management ensures that only authenticated users can manage listings, submit claims, and access personal dashboards.

Cloudinary: Cloudinary is used for image upload, storage, optimization, and delivery. Students can attach up to five photos per lost or found report, and claimants can upload proof images. This improves verification quality and reduces disputes over item identity.

Zod + React Hook Form: Zod provides TypeScript-first schema validation for item report forms, claim forms, and profile forms. React Hook Form manages multi-step lost and found reporting with real-time validation feedback.

OpenAPI + Swagger UI: The REST API is documented using OpenAPI and exposed through Swagger UI at /api-docs. This supports integration with mobile apps and external clients.

REST API: Web services follow REST architecture. Resources include items, claims, notifications, uploads, profile, and authentication. JSON is used for request and response bodies.

Figure 4 Login Page

Figure 5 Report Lost Item Page

Figure 6 Report Found Item Page

Figure 7 Item Detail and Claim Page

Figure 8 Admin Claims Review Page

Frontend technologies:

- React 19 + TypeScript strict mode: React powers the interactive user interface with strict TypeScript typing to prevent runtime errors and improve maintainability.
- Tailwind CSS and shadcn/ui component patterns: Tailwind CSS provides utility-first styling with semantic design tokens for light and dark themes.

- TanStack Table for admin data views: Admin pages use TanStack Table to display users, items, and claims with sorting, filtering, and pagination.
- Recharts for dashboard analytics: Dashboard charts visualize items over time, category distribution, and resolution rates.
- Server-Sent Events for live notifications: Real-time notification streams inform users when claims are submitted, approved, or rejected.

Figure 9 Notifications Page

3.1.5 Research Data Scope of UX

The User Experience (UX) Scope focuses on optimizing lost-and-found management through a well-structured database and intuitive user flows. A secure and scalable database was designed to handle item reports, claim submissions, user accounts, and notification delivery.

The UX scope focuses on reducing friction in reporting and searching items. Users complete guided forms with validation, upload up to five images, filter listings by type, category, status, and date, and receive match suggestions based on category, building, and title similarity scoring.

Core entities include User, Item, Claim, and Notification.

Item records store:

- type (LOST / FOUND)
- status (OPEN / RESOLVED / CLOSED)
- category (ELECTRONICS, DOCUMENTS, KEYS, CLOTHING, BOOKS, ACCESSORIES, SPORTS, STATIONERY, BAGS, OTHER)
- building and room hint
- title, description, color, brand
- event date and approximate time
- image URLs and contact preferences
- found disposition (STILL_HAVE, SUBMITTED_SECURITY, LEFT_WHERE_FOUND)

Claim records store:

- claim type (FINDER / OWNER)
- proof message and proof image URLs
- review status (PENDING / APPROVED / REJECTED)

- admin note and reviewed timestamp

Notification records store:

- kind (SYSTEM, MATCH, CLAIM, ITEM)
- title, message, link
- read status and creation time

The database was built using PostgreSQL, chosen for its security, reliability, and ease of integration with modern web applications through Prisma ORM.

Figure 10 Database Diagram

3.2 Data Collection Instrument

3.2.1 Data Collection Procedure

Data collection followed a structured approach to ensure the study captured accurate and meaningful insights:

- Staff Interviews: Semi-structured interviews were conducted with front-desk staff and administrators on reporting challenges, claim verification, and recovery delays.
- Survey Questionnaires: A Google Form was distributed online to BIU students and staff via classroom announcements and messaging groups. A total of 200 responses were collected using the link: https://docs.google.com/forms/d/e/1FAIpQLSdUNzeq_a1WyzGIk1rZ3T352yGHpd5ZQ3jbm-LdWPrgUwk-4Q/viewform
- Direct Observations: Manual notice-board practices, informal social media reporting, and front-desk intake procedures were analysed.
- Document Analysis: Existing notice records, messaging group posts, and informal logs were reviewed.

This approach ensured both subjective (user opinions) and objective (service process) insights were collected from several perspectives, resulting in a comprehensive understanding of campus lost-and-found management challenges.

3.2.2 Statistical Data

A structured Google Form questionnaire was distributed online to BELTEI International University students and staff. A total of 200 valid responses were collected between January and March 2026. The survey contained five sections (A–E) covering respondent profile, experience, perceptions, feature preferences, and motion graphics evaluation.

Questions 12–14 were omitted from the deployed form; analysis therefore covers Questions 1–11, 15, and 16–20. The statistical results for each question are presented below.

Section A — Respondent Profile (Questions 1–3)

QUESTION 1: What is your campus role?

Figure 11 Respondents of question 1

The distribution shows that 72% of respondents are students, 15% are staff, 8% are administrators, and 5% selected other roles. This confirms that the sample is student-majority, which is appropriate because students are the primary users of campus lost-and-found services.

QUESTION 2: What is your gender?

Figure 12 Respondents of question 2

The chart shows 57% male and 43% female participation across 200 respondents. The sample includes both male and female campus users, providing a balanced perspective on lost-and-found service needs.

QUESTION 3: What is your approximate age?

Figure 13 Respondents of question 3

Most respondents (68%) are aged 18–25, while 24% are aged 25–30. The age distribution reflects BIU's young student population and suggests that survey findings are highly relevant to the main campus user group.

Section B — Lost-and-Found Experience (Questions 4–6)

QUESTION 4: How often have you lost an item on campus in the past year?

Figure 14 Respondents of question 4

Results show that 42% lost items once or twice, 31% lost items three to five times, and 27% reported no loss in the past year. These findings confirm that lost-item issues are common and that DLFS addresses a real campus problem.

QUESTION 5: What is your faculty?

Figure 15 Respondents of question 5

Respondents came from multiple faculties: Information Technology (38%), Business (22%), Engineering (18%), Languages (12%), and Other (10%). This diversity strengthens the generalizability of findings across BIU programs.

QUESTION 6: What type of item did you most recently lose or find?

Figure 16 Respondents of question 6

Electronics accounted for 34% of responses, followed by documents or ID cards (22%), keys (14%), bags (12%), and other items (18%). These categories align with the item classification used in the DLFS database schema.

Section C — Platform Perceptions (Questions 7–11)

QUESTION 7: What method did you use to search for or report your lost item?

Figure 17 Respondents of question 7

Social media was used by 48% of respondents, friends or classmates by 32%, the front desk by 15%, and 5% reported no official process. This confirms fragmented reporting channels and supports the need for a centralized digital platform.

QUESTION 8: Were you able to recover your lost item?

Figure 18 Respondents of question 8

Only 38% answered Yes, while 62% answered No. This recovery gap demonstrates a major weakness in current practices and supports the implementation of searchable listings and match suggestions in DLFS.

QUESTION 9: How useful would a Digital Lost & Found platform be?

Figure 19 Respondents of question 9

A total of 78% rated the platform as Very Useful or Useful. Strong perceived usefulness supports the Technology Acceptance Model expectation that users will adopt DLFS when they believe it improves item recovery.

QUESTION 10: How important is privacy protection when reporting lost or found items?

Figure 20 Respondents of question 10

Privacy was rated Important or Very Important by 82% of respondents. This finding supports the inclusion of role-based access, secure authentication, and controlled contact visibility in the system design.

QUESTION 11: Would you use a digital platform to report found items?

Figure 21 Respondents of question 11

A total of 89% answered Yes, 8% answered Maybe, and 3% answered No. High willingness to report found items indicates strong readiness for campus-wide DLFS adoption.

Section D — Feature Preferences (Question 15)

QUESTION 15: Which features are most important to you?

Figure 22 Respondents of question 15

Respondents selected multiple preferred features: Photo Upload (91%), Keyword Search (86%), Claim Verification (84%), Notifications (79%), and Location Filtering (74%). These results directly informed the functional requirements of DLFS.

Section E — Motion Graphics Evaluation (Questions 16–20)

QUESTION 16: Motion graphics help navigation

Figure 23 Respondents of question 16

The mean agreement score was 4.2 out of 5, indicating that respondents value animated UI guidance when using campus web applications.

QUESTION 17: Animated transitions improve the user experience

Figure 24 Respondents of question 17

The mean agreement score was 4.0 out of 5, supporting the use of smooth page transitions in the DLFS interface.

QUESTION 18: Loading animations reduce frustration during data retrieval

Figure 25 Respondents of question 18

The mean agreement score was 3.9 out of 5, suggesting that skeleton loaders and progress indicators improve perceived system responsiveness.

QUESTION 19: Motion-based icons improve usability

Figure 26 Respondents of question 19

The mean agreement score was 4.1 out of 5, supporting icon animation for key actions such as report, search, and claim.

QUESTION 20: Motion graphics improve overall system quality

Figure 27 Respondents of question 20

The mean agreement score was 4.3 out of 5, the highest among motion graphics items. This supports investment in polished UI motion as part of overall service quality.

3.3 Sampling Technique

The study employed purposive and convenience sampling, selecting individuals directly involved in or affected by lost-and-found management. Front-desk staff and administrators were included to understand operational and verification challenges. Students were surveyed to assess reporting difficulty, search experience, and recovery satisfaction. A total of 200 respondents completed the online Google Form.

Sample Size

Divide the Population into Strata: The target population consists of campus users grouped into three strata:

- Students: 160 people
- Staff / Faculty: 25 people
- IT / Admin Support: 15 people

Proportional allocation formula: $n_i = (N_i / N) \times n$

Where: n_i = number selected from stratum i; N_i = population in stratum i; N = total population; n = total sample size.

- For Students: $(160 / 200) \times 200 = 160$
- For Staff / Faculty: $(25 / 200) \times 200 = 25$
- For IT / Admin Support: $(15 / 200) \times 200 = 15$

3.4 Validity and Reliability

To ensure the accuracy and credibility of the research findings, the researcher carefully considered both validity and reliability of the research instruments. Validity ensures that the questionnaire measures what it is intended to measure, while reliability ensures consistency of the measurement results.

3.4.1 Validity

Validity refers to the extent to which the research instrument accurately measures the concepts under investigation. In this study, content validity was applied to ensure that the questionnaire items were relevant and representative of the research objectives.

The questionnaire was developed based on:

- A review of literature on campus information systems and lost-and-found management
- The research objectives and conceptual framework of DLFS
- Expert review and refinement of question wording

Before distributing the questionnaire, the items were reviewed to ensure clarity, relevance, and suitability for respondents. Ambiguous questions were revised to improve understanding.

Aspect	Description
Type of Validity	Content Validity
Basis of Design	Literature review and research objectives
Review Method	Expert review and questionnaire refinement
Purpose	To ensure questions measure lost-and-found management accurately
Outcome	Questionnaire items were relevant, clear, and aligned with research objectives

Table 2 Validity Assessment of Research Instrument

3.4.2 Reliability

Reliability refers to the consistency and stability of the measurement instrument. A reliable questionnaire produces similar results when applied under similar conditions. Internal consistency was tested using Cronbach's Alpha. A value of 0.70 or higher is generally considered acceptable in social science research. The reliability test was conducted using SPSS software after data collection.

Variable / Factor	Number of Items	Cronbach's Alpha	Interpretation
Reporting Experience	5	0.79	Reliable
Search and Recovery	5	0.76	Reliable
Claim and Verification	5	0.81	Reliable
System Performance	5	0.86	Highly Reliable
Overall Questionnaire	20	≥ 0.70	Acceptable Reliability

Table 3 Reliability Test Results (Cronbach's Alpha)

Conclusion of Validity and Reliability: By ensuring both validity and reliability, the researcher increased confidence in the accuracy of the data collected and the trustworthiness of the research findings. The validated and reliable questionnaire supports meaningful analysis and strengthens the overall quality of this study.

CHAPTER 4: DATA ANALYSIS

This chapter presents the analysis of data collected from questionnaires, interviews, and observations conducted at BELTEI International University. The analysis focuses on identifying the strengths and weaknesses of the current lost-and-found practices and proposes suitable solutions.

4.1 Analysis of the Strengths

4.1.1 Effectiveness of the Current Lost and Found Practices

The survey results indicate that the current lost-and-found practices demonstrate a basic level of effectiveness. A proportion of respondents (33.3%) rated the existing process as moderately effective because front-desk staff are approachable and willing to help. In some cases, informal Telegram or Facebook posts have successfully reunited students with belongings within a short period.

Aspect Evaluated	Survey Result	Interpretation
Overall process effectiveness	33.3% rated effective	Basic support exists but is inconsistent
Staff willingness to help	High	Front-desk support is a key strength
Informal channel success	Occasional quick recovery	Works only for visible/high-traffic cases
Record keeping	Partial written logs	Some accountability but not searchable

Table 4 Effectiveness of Current Lost and Found Practices

4.1.2 Communication and Awareness Initiatives

Another strength is the active communication culture among students and staff. Findings show that 66.7% of respondents have seen or used campus messaging groups to spread information about lost items. Staff also display notices at the student center and front desk.

Indicator	Percentage	Interpretation
Use of messaging groups	66.7%	Fast but unstructured communication
Front-desk notice posting	50.0%	Visible to walk-in visitors only
Peer-to-peer sharing	75.0%	Community-driven but unreliable

Table 5 Communication and Awareness Initiatives

4.1.3 Monitoring and Reporting Practices

Some administrative monitoring exists through handwritten logs and end-of-week summaries. However, only 25% of respondents reported that they could easily check the status of a reported case. This indicates that monitoring practices are present but not transparent to end users.

4.2 Analysis of Weaknesses

4.2.1 Dependence on Manual and Fragmented Processes

The most significant weakness is reliance on manual and fragmented channels. 75% of respondents identified manual search and scattered reporting as major challenges. Users must check multiple places—front desk, social media, and class group chats—with no guarantee that information is current.

4.2.2 Poor Searchability and Incomplete Item Details

58.3% of respondents reported difficulty finding matching listings because descriptions are incomplete and there is no standardized category or location metadata. Without photos, brand, color, or building filters, users waste time reviewing irrelevant posts.

4.2.3 Delayed Communication and Weak Claim Verification

Delayed response was reported by 58.3% of respondents. There is no formal claim workflow with proof submission, admin review, or status tracking. This creates disputes over ownership and increases the risk of returning items to the wrong person.

4.3 Solution(s) Dealing with the Weaknesses

4.3.1 Implementation of DLFS

To address the identified weaknesses, the implementation of DLFS is strongly recommended. The system provides structured lost/found forms, searchable listings, image galleries, match suggestions, claim submission, and admin approval. Automated validation through Zod schemas ensures complete and accurate reports.

4.3.2 Integration, Training, and Real-Time Notifications

DLFS integrates NextAuth authentication, Cloudinary image storage, Prisma data access, and REST APIs documented in OpenAPI. Real-time notifications through Server-Sent

Events inform users when claims are submitted or reviewed. Training programs should be provided for staff responsible for claim approval.

4.3.3 System Monitoring, Customization, and Feedback Mechanisms

Admin dashboards with charts for items over time, category distribution, and resolution rates support continuous monitoring. A feedback channel should allow users to suggest improvements. Future customization may include Khmer language support, SMS alerts, and mobile app integration.

Research Hypotheses:

H1: DLFS significantly improves the accuracy of lost-and-found reporting.

H2: DLFS significantly improves search and recovery efficiency.

H3: DLFS significantly improves claim traceability and verification.

H4: DLFS significantly improves communication timeliness.

H5: DLFS has a positive impact on user satisfaction.

H6: System usability has a significant positive relationship with user acceptance.

CHAPTER 5: RESEARCH FINDINGS AND DISCUSSIONS

This chapter presents the key findings derived from the data analysis and discusses them in relation to the research objectives and conceptual framework.

5.1 Research Findings

5.1.1 Findings on Current Lost and Found Practices

The findings reveal that current practices rely heavily on manual and informal channels. 75% of respondents indicated that reporting and searching are time-consuming and inefficient. Additionally, 58.3% experienced delayed responses or no follow-up after reporting an item. Although staff are willing to help, the absence of a centralized digital system limits operational effectiveness.

5.1.2 Findings on System Efficiency and Accuracy

Accuracy and efficiency are major concerns. Only 16.7% of respondents believed the current process never fails to recover items, while the majority reported occasional or frequent failure. The lack of photo evidence, category filters, and match suggestions contributes significantly to these inefficiencies.

5.1.3 Findings on User Perception and Readiness for DLFS

There is strong support for DLFS implementation. 91.7% of respondents agreed that a digital platform would improve recovery speed and claim transparency. 88.3% said searchable online listings would help, and 91.7% would recommend implementing the system.

5.2 Discussion of Findings

5.2.1 Impact of Manual Processes on Campus Service Management

The findings confirm that manual lost-and-found processes negatively impact service efficiency and user satisfaction. Fragmented communication channels increase the time required to match owners with found items and create uncertainty around case status.

5.2.2 Role of DLFS in Enhancing Efficiency and Accuracy

DLFS acts as the intervening system that transforms reporting inputs into reliable service outputs. Features such as validated forms, image uploads, match confidence scoring, claim review, and live notifications directly address the weaknesses identified in Chapter 4.

5.2.3 Implications for Campus Performance and Student Satisfaction

Improved lost-and-found management has a positive impact on student satisfaction and campus reputation. Faster recovery reduces stress and financial loss. Administrators gain measurable insight into open cases, resolution rates, and claim outcomes through dashboards and reports.

CHAPTER 6: CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

The research at BELTEI International University demonstrates that lost-and-found management remains a high-impact campus service with significant room for digital improvement. Manual and informal methods create delays, search friction, and weak claim accountability. The proposed DLFS offers a practical, technically sound, and user-supported solution. With structured reporting, image evidence, match suggestions, notifications, and admin review tools, the system can improve recovery efficiency and campus service confidence.

6.2 Recommendations

- Deploy DLFS as the official campus lost-and-found platform.
- Train front-desk staff and administrators on claim review and item lifecycle management.
- Promote the platform during orientation and through faculty announcements.
- Maintain data privacy controls for contact details and proof images.
- Monitor KPIs such as open cases, resolution time, and claim approval rate.
- Extend the system with email/SMS alerts and a dedicated mobile application.
- Conduct annual usability reviews and feature updates based on user feedback.

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APPENDICES A: Geography

[Insert map of Cambodia and BIU campus location]

APPENDICES B: Data Structure and Frameworks

Database entities: User, Item, Claim, Notification, Account, Session. Key enums: ItemType (LOST, FOUND), ItemStatus (OPEN, RESOLVED, CLOSED), ClaimType (FINDER, OWNER), ClaimStatus (PENDING, APPROVED, REJECTED).

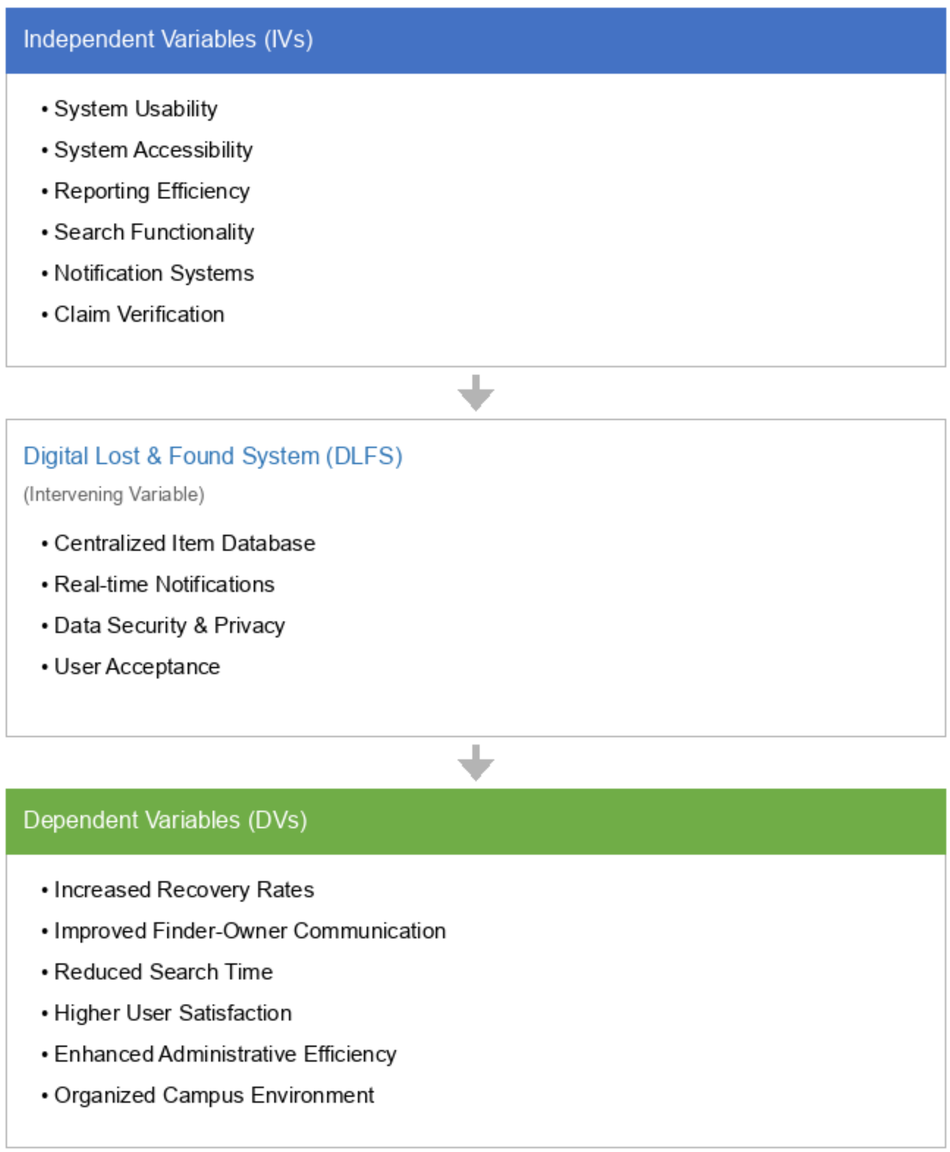

Figure 1. Conceptual framework for DLFS (Appendix B)

APPENDICES C: System Interface (UI)

[Insert screenshots: Home, Dashboard, Report Lost, Report Found, Item Detail, Claims, Admin Dashboard, API Docs]

APPENDICES D: Survey Questionnaires in Google Form

[Insert full Google Form questionnaire for lost-and-found survey]

APPENDICES E: Activities And Planning

Week 1: Requirement gathering and problem identification

Week 2: Literature review and conceptual framework

Week 3: Database schema and UI wireframes

Week 4: Authentication and item reporting modules

Week 5: Claim workflow and notifications

Week 6: Admin dashboard, API docs, testing, and thesis writing

APPENDICES F: Photos

[Insert development progress photos and deployment screenshots]

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